

— NCERT · DEPFE · CAREER EXHIBITION

# Artificial Intelligence Engineer

A career talk for senior secondary students — the pathways, the work, and the future of building intelligent systems.

AUDIENCE

**Classes IX – XII**

VENUE

**KV RK Puram, Sector-2**

DEPARTMENT

**DEPFE, NCERT, New Delhi**

— WHAT WE WILL COVER

# Ten questions about a career in AI

A structured walk-through of everything you need to decide whether this profession is right for you.

**01 Educational Pathway**

Subjects, exams and degrees

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**03 Working Conditions**

Hours, teams and environment

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**05 Career Progression**

Entry to leadership roles

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**07 Employment Opportunities**

Sectors and geographies

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**09 Courses & Internships**

Building employability

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**02 Nature of Work**

Day-to-day tasks and tools

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**04 Personal Attributes**

Skills and dispositions

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**06 Estimated Salary**

Pay at each stage

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**08 Best Colleges in India**

Where to study

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**10 Future Outlook**

Long-term relevance

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# How you get into the field

## ■ School (Class XI–XII)

- **Science stream with Mathematics** — Maths is essential
- Physics & Chemistry (PCM) for engineering routes
- Opt for **Computer Science / Informatics** if offered
- Start coding early — Python is a great first language

## ■ Entrance Exams

- **JEE Main & Advanced** — IITs, NITs, IIITs
- **BITSAT** — BITS Pilani campuses
- **CUET** — central universities
- State CETs & university-specific tests

## ■ Degree Routes

- **B.Tech / B.E.** in CSE, AI & ML, or Data Science
- **B.Sc** in Computer Science, Maths or Statistics
- **M.Tech / M.S. / PhD** for research & senior roles
- Online degrees & a strong project portfolio also count

- **Key point:** There is no single fixed route. A solid grounding in mathematics and programming matters more than any one degree.

# What an AI engineer actually does

## ■ Core Responsibilities

- Design, train and fine-tune **machine learning & deep learning** models
- Collect, clean and prepare data; engineer features
- Deploy models into products and **monitor them in production (MLOps)**
- Work on NLP, computer vision, recommendation & large language models
- Write reliable, production-grade code and run experiments

## ■ Tools & Technologies

- **Languages:** Python, SQL, sometimes C++ / Java
- **Frameworks:** PyTorch, TensorFlow, scikit-learn, Hugging Face
- **Cloud & DevOps:** AWS, GCP, Azure, Docker, Git
- **Data:** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product & software teams

# The work environment

## ■ Hours & Arrangements

- Typical **40–45 hour** work weeks
- Extra hours near product launches or deadlines
- **Hybrid and remote** work are common in tech
- On-site work for robotics, hardware or defence projects

## ■ Teams & Culture

- Small **cross-functional teams** (5–10 people)
- Agile sprints, code reviews and stand-ups
- Global collaboration across time zones
- High screen time and a constant need to keep learning

# What helps you succeed

## ■ Technical Aptitude

- Strong **mathematics** — linear algebra, probability, statistics
- Logical reasoning & programming aptitude
- Comfort with abstraction and problem solving

## ■ Mindset

- **Curiosity** and self-driven learning
- Patience for experimentation and debugging
- Attention to detail and rigour with data

## ■ Soft Skills

- Explaining complex ideas **simply**
- Teamwork and collaboration
- Ethical judgement and responsibility

# From entry level to leadership

## **Intern / Trainee**

Starting out

Learn on real projects, assist with data and experiments while completing studies.

## **Junior AI / ML Engineer**

0–2 years

Build and test models under guidance; strengthen coding and data skills.

## **AI / ML Engineer**

2–5 years

Own features end-to-end, deploy models to production, mentor juniors.

## **Senior AI Engineer**

5–8 years

Lead complex systems, set technical direction, review architecture.

## **Lead / Principal / ML Architect**

8–12 years

Design large-scale AI systems and guide multiple teams.

## **Head of AI / Director / CTO**

12+ years

Own AI strategy and organisation-wide impact.

## Indicative compensation in India

### Entry / Junior 0–2 years

First roles; varies widely by employer and city.

₹6–12 LPA

### Mid-level 2–5 years

Proven delivery and production experience.

₹12–25 LPA

### Senior 5–8 years

Technical ownership and mentoring.

₹25–45 LPA

### Lead / Principal 8–12 years

Architecture and multi-team leadership.

₹45–80 LPA

### Director / Head of AI 12+ years

Strategy; top product firms pay well above.

₹80 LPA+

*Figures are broad, indicative ranges (annual, in Indian Rupees) and vary significantly with company, location, skills and global roles. Top product companies and overseas positions can pay considerably more.*

## Where the demand is

### ■ Sectors

- IT & software, e-commerce
- Fintech, healthcare, automotive
- Defence, agriculture, telecom
- Edtech, media & entertainment

### ■ Organisations

- Global: Google, Microsoft, Amazon, NVIDIA, Meta
- Indian IT: TCS, Infosys, Wipro
- Startups: Sarvam AI, Krutrim & more
- Research: ISRO, DRDO, IITs, IISc

### ■ Geographies

- Bengaluru, Hyderabad, Pune
- Gurugram, NCR, Chennai
- Global: US, Europe, Singapore, UAE
- Remote, freelance & consulting

## Recommended institutions

### ■ Premier Engineering Institutes

- **IITs** — Bombay, Delhi, Madras, Kanpur, Kharagpur, Hyderabad (dedicated AI department)
- **IISc Bangalore** — leading for AI research
- **NITs** — Trichy, Surathkal, Warangal
- **BITS Pilani** — Pilani, Goa, Hyderabad

### ■ Strong CS & AI Programmes

- **IIITs** — Hyderabad, Delhi, Bangalore
- Delhi University (DU) & DTU / NSUT
- VIT, Manipal, Ashoka University
- Reputed state & private universities with AI/DS courses

# Strengthening employability

## ■ Certifications

- DeepLearning.AI specialisations (Andrew Ng)
- Google, AWS & Azure ML certifications
- NVIDIA Deep Learning Institute

## ■ Courses & Practice

- Coursera / fast.ai machine learning courses
- Stanford CS231n (vision), CS224n (NLP)
- **Kaggle** competitions & Kaggle Learn

## ■ Internships & Portfolio

- Research internships at IITs / IISc
- Google Summer of Code, industry internships
- Open-source & **GitHub project portfolio**

# The long-term relevance of AI

## ■ Emerging Trends

- **Generative AI, agents and multimodal models** are reshaping every industry
- Government push through the **IndiaAI Mission**
- Rapid adoption across healthcare, finance, agriculture and governance

## ■ What it means for you

- One of the **fastest-growing, highest-demand** careers
- Rising focus on **responsible & ethical AI** and safety
- Lifelong learning is essential — the field evolves quickly

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## IN SUMMARY

# Build the future — thoughtfully.

Start with strong mathematics and coding, choose a pathway that fits you, keep building projects, and never stop learning. AI rewards curiosity and persistence.

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Thank you · Questions & Discussion